



Software project Report On

Airline Reservation System

Submitted to



Savitribai Phule Pune University

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Computer Application

CERTIFICATE



Savitribai Phule Pune University

This is to certify that, **Miss: Mehraj Muktar Shaikh Class- TYBBA-CA ,**
Roll No. 3615 has satisfactorily completed the project under the subject
“Software Project-II ” having the title **“ Arline Reservation System”**. As
laid down by the Savitribai Phule Pune University during the year
2025-2026

Project Guide

Internal Examiner

Head of Department

External Examiner



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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide for their valuable guidance, continuous support, and encouragement during the development of the **Airline Reservation System** project. Their suggestions, knowledge, and motivation helped me understand the project concepts clearly and complete the work successfully.

I am also thankful to the faculty members of my college for providing the necessary knowledge, resources, and a supportive learning environment throughout my academic studies. Their guidance played an important role in the successful completion of this project.

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Finally, I would like to express my heartfelt thanks to my parents and family members for their constant encouragement, patience, and moral support. Their motivation inspired me to work hard and complete this project successfully.

1.ABSTRACT

The Airline Reservation System (ARS) is a computerized system developed to automate the process of booking airline tickets and managing flight-related information. It provides an efficient platform for passengers and airline staff to perform reservation activities such as searching available flights, checking seat availability, booking tickets, making payments. The system maintains a centralized database that stores important information including flight schedules, passenger details, ticket status, and booking records.

The main objective of the Airline Reservation System is to simplify the reservation process and reduce the time and effort required for manual booking. Passengers can easily view flight details, compare available options, and reserve seats according to their preferences. The system also allows users to modify or cancel their bookings when necessary. For airline management, it helps in maintaining accurate records of flights, passengers, and seat availability.

The system is designed with different modules such as flight management, reservation management, passenger information management, and payment processing. These modules work together to ensure smooth functioning of the airline booking process. Airline staff can update flight schedules, manage seat allocation, and generate reports related to reservations and passenger information.

2.Introduction

An Airline Reservation System (ARS) is a web-based or desktop application that automates flight booking, seat inventory management, and ticketing processes. It allows passengers to search, book, and pay for flights online while enabling staff to manage schedules and passenger records efficiently. These systems integrate with Global Distribution Systems (GDS) to facilitate real-time, global ticket distribution, significantly reducing manual effort and improving operational efficiency. An Airline Reservation System (ARS) is a computerized system used to manage flight bookings, schedules, passenger information, and ticket reservations. It helps airlines and travel agencies handle the process of booking airline tickets quickly and efficiently. The system allows passengers to check flight availability, select seats, book tickets, and make payments online without visiting the airline office.

2.1 Motivation

The motivation for an Airline Reservation System (ARS) is to modernize, automate, and centralize the inefficient, manual process of booking flights, resulting in improved efficiency, increased sales, and better customer service. These systems provide 24/7 real-time access to scheduling and inventory management, ensuring accuracy while reducing operational costs for airlines.

The Airline Reservation System is to improve the efficiency and convenience of the flight booking process. In the traditional system, ticket booking and management were done manually, which was time-consuming and prone to errors.

2.2 Problem Statement

The problem is to develop a centralized system that allows passengers to search for available flights, book tickets, cancel reservations, and view booking details easily. At the same time, the system should help airline staff manage flight schedules, passenger information, seat availability, and payment records efficiently.

Therefore, there is a need to develop an Airline Reservation System that can automate the process of searching flights, booking tickets, managing passenger details, and handling cancellations or updates. The system should provide a user-friendly interface for customers and administrators, improve data accuracy, reduce manual workload, and ensure quick and secure management of airline reservation information.

2.3 Objectives and Goals

- Automate ticket booking and reservation process.
- Manage passenger and flight information efficiently.
- Provide quick booking and modification of tickets.
- Maintain secure and organized data records.

2.4 Literature Survey

This literature survey reviews Airline Reservation Systems (ARS), highlighting their evolution from manual, error-prone processes to computerized, web-based systems that enhance efficiency, security, and user experience. Modern ARS platforms enable real-time flight

searches supported by secure relational databases like MySQL user-friendly interfaces.

2.5 Scope and Limitation

Scope:

- Flight Search and Booking.
- Passenger and Inventory Management.
- Payment Gateway Integration.

Limitation:

- Internet Dependence.
- Security Risks.
- Limited Customization.

3.SYSTEM ANALYSIS

3.1 Existing System and Limitations

Existing System:

- Passengers can book, cancel, or modify tickets through the system.
- Payment is done using online payment methods.
- The system stores passenger and flight details in a database.

Limitations:

- Sometimes the system becomes slow due to heavy traffic.
- Internet connection is required to access the system.
- Technical errors may cause booking failures.

3.2 Project Perspective and Features

- User Registration and Login – Passengers can create an account and login.
- Flight Search – Users can search flights by date, destination, and source.
- Ticket Booking – Passengers can book tickets online.
- Online Payment – Payment can be made through digital methods.
- Admin Management – Admin can add, update, or delete flight details.

- Booking History – Passengers can view their previous bookings.

3.3 Stakeholders

Stakeholders are the people or groups who are involved in or affected by the Airline Reservation System.

1. Passengers – Use the system to search flights, book tickets, and manage reservations.
2. Airline Management – Monitor airline operations, bookings, and overall system performance.
3. Airline Staff – Handle flight schedules, passenger records, and ticket management.
4. Travel Agents – Use the system to book tickets for customers.
5. System Administrator – Maintains the system, manages databases, and ensures security.
6. Payment Service Providers – Process online payments and transactions for ticket booking.

Requirement Analysis

3.4 Functional Requirements

- User Registration – The system should allow new users to create an account.
- User Login – The system should allow registered users to login securely.
- Search Flights – Users should be able to search flights by source, destination, and date.

- View Flight Details – The system should display flight schedule, price, and seat availability.
- Book Ticket – Users should be able to book flight tickets online.
- Online Payment – The system should process payments through online methods.
- Booking History – Users should be able to view their previous bookings.
- Admin Management – Admin should be able to add, update, or delete flight information.

3.4 Performance Requirements

- The system should respond quickly to user requests.
- The system should handle many users at the same time.
- Flight search results should be displayed within a few seconds.
- The system should process ticket booking and payment efficiently.
- Quick Database Access.

3.5 Security Requirements

- User Authentication – The system require username and password for login.
- Data Protection – Passenger and payment information should be stored securely.
- Secure Payment Processing: Online payment transactions should be encrypted to ensure safe financial transactions.

4.SYSTEM DESIGN

The system design of an Airline Reservation System describes the overall structure of the system, including its components, database, and how users interact with it. The system is designed to manage flight information, passenger details, ticket booking, and payment processes efficiently

4.1 Main Modules of the System

- User Module: Allows passengers to register, log in, search flights, book tickets, and view booking details.
- Flight Management Module: Used by administrators to add, update, or delete flight schedules and seat availability.
- Reservation Module: Handles ticket booking and reservation confirmation.
- Payment Module: Manages online payments and transaction records.
- Report Module: Generates reports for bookings, passengers, and flight schedules.

4.2 User Interface Design

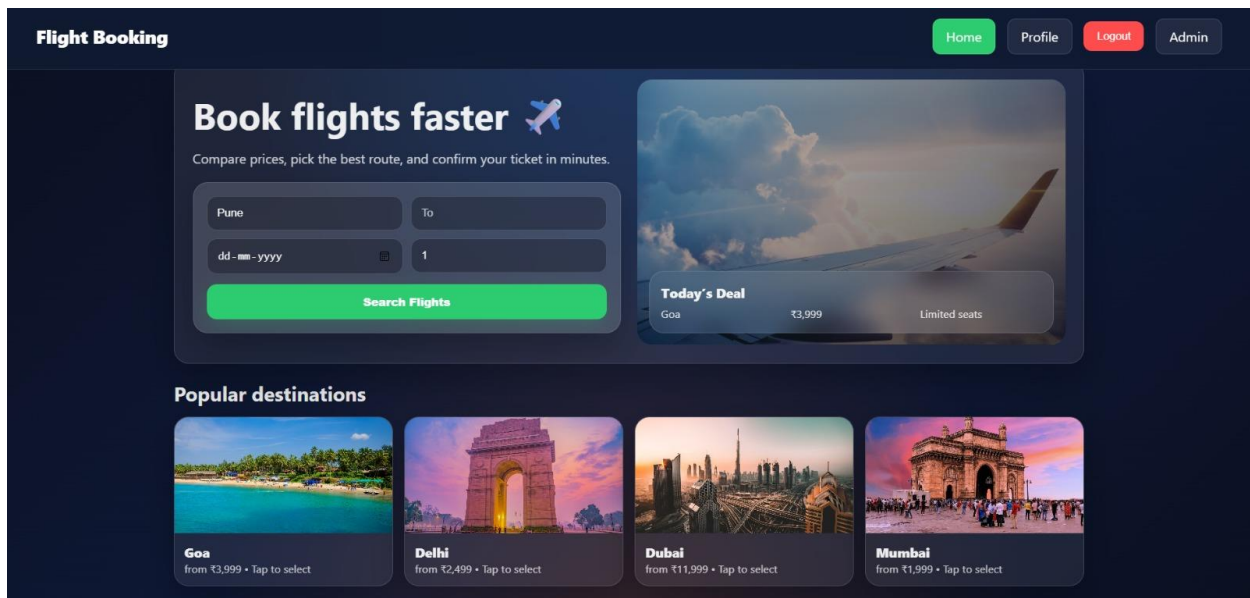
The system provides a simple and user-friendly interface where users can easily search flights, enter passenger details, and complete reservations.

Overall, the system design ensures that the Airline Reservation System works smoothly, manages large amounts of data, and provides fast and reliable service to passengers.

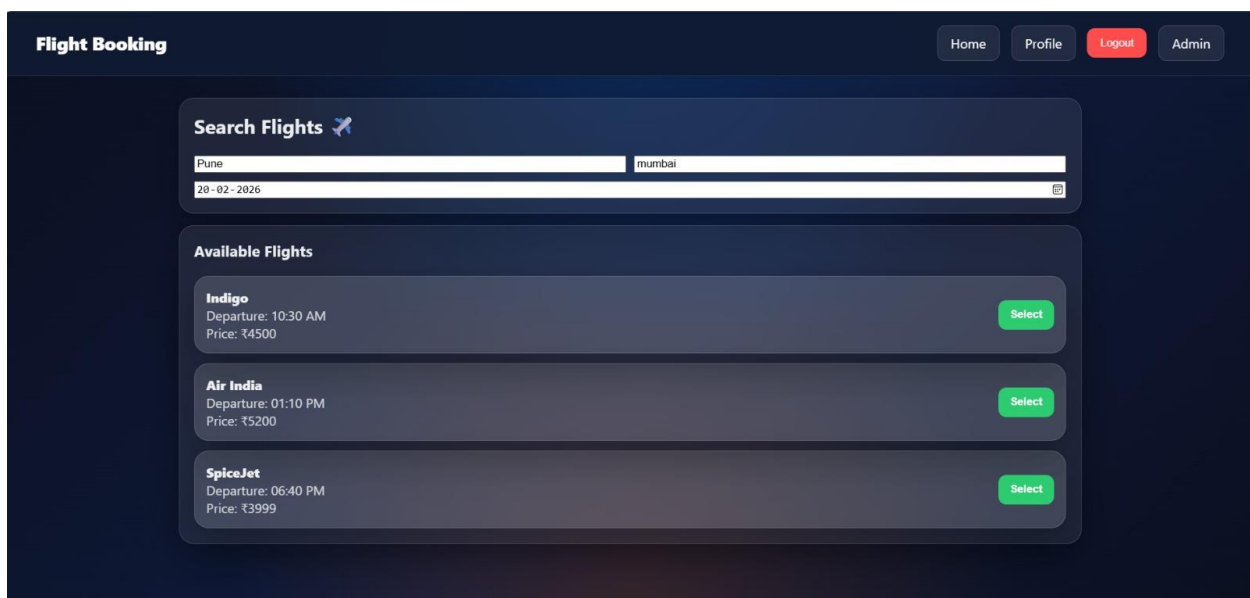
4.5 User Interfaces

The interface includes features such as login and registration pages, flight search options, booking forms, payment pages, and ticket confirmation screens. Passengers can easily enter travel details like departure city, destination, and travel date to find available flights.

1.Home page,Fight Booking page



2.Fight Booking page



Flight Booking

Home

Profile

Logout

Admin

Flight Summary ✈️

From: Pune

To: mumbai

Date: 2026-02-20

Airline: Indigo

Departure Time: 10:30 AM

Price: ₹4500

Passenger Details

pranav pote

Age

Male

Logged in as: pranavpote2933@gmail.com

Confirm Booking

Flight Booking

Home

Profile

Logout

Admin

✏️ Ticket Confirmed

Booking ID: 10

Name: pranav pote

Age: 20

Gender: Male

From: Pune

To: mumbai

Date: 2026-02-20

Airline: Indigo

Departure: 10:30 AM

Price: ₹4500

User: pranav pote

Email: pranavpote2933@gmail.com

Proceed to Payment

Payment:

Flight Booking

HomeProfileLogoutAdmin

Payment

Ticket Summary

Booking ID: 10
Name: pranav pote
From: Pune
To: mumbai
Date: 2026-02-20
Airline: Indigo
Departure: 10:30 AM
Total Price: ₹4500

Select Payment Method

UPI

Enter UPI ID

Pay Now

Admin Panel:

Flight Booking

HomeProfileLogoutAdmin

Admin Panel

All bookings, users, passenger details, flight info and payment status.

Filters

Search by user, email, city, airline, booking id...

All Status

All Payment

Export CSV

Booking #10

CancelDelete

User ID: 12
User Name: pranav pote
Email: pranavpote2933@gmail.com

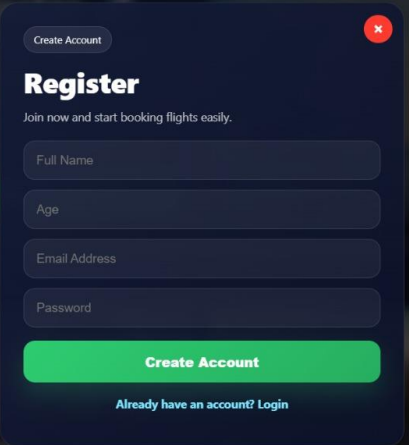
Passenger: pranav pote
Age: 20
Gender: Male

Airline: Indigo
From: Pune

5.Login/Register Pages:

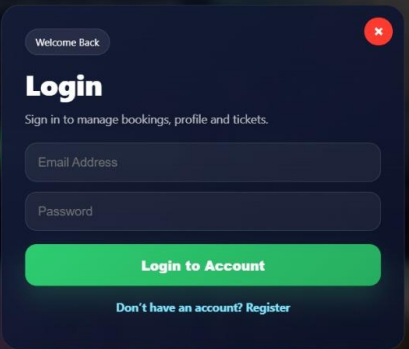
- Login Page: Users enter username and password to sign in.
- Register Page: New users enter name, address, password, and contact details.

➤ Register page:



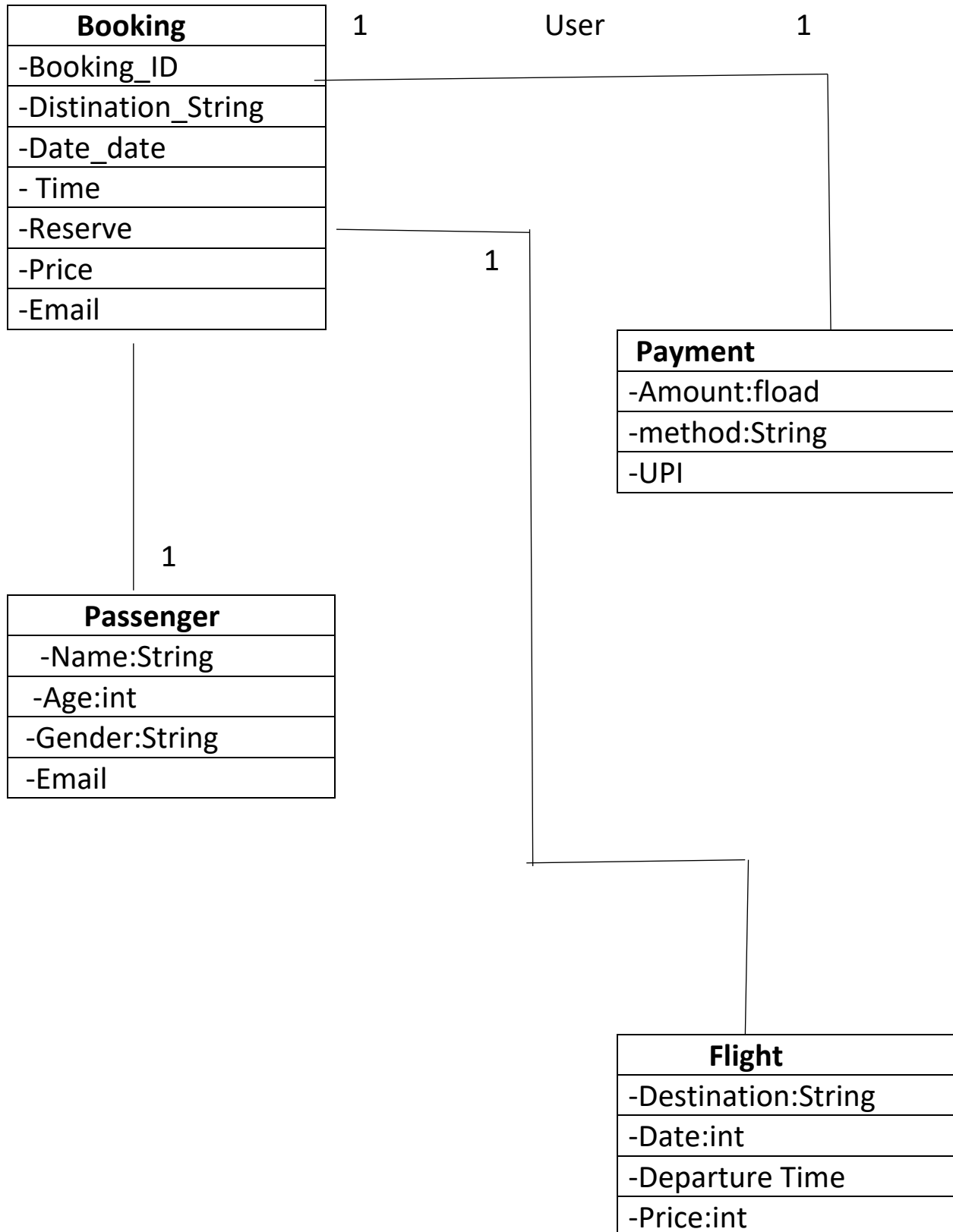
The image shows a dark-themed web application with a modal overlay for registration. The modal has a title bar with a 'Create Account' button and a close icon. The main heading is 'Register', followed by the subtext 'Join now and start booking flights easily.' Below this are four input fields: 'Full Name', 'Age', 'Email Address', and 'Password'. A prominent green button labeled 'Create Account' is at the bottom, with a link 'Already have an account? Login' underneath it. The background is a blurred view of the main application interface.

➤ Login page:



The image shows the same dark-themed web application with a modal overlay for login. The modal has a title bar with a 'Welcome Back' button and a close icon. The main heading is 'Login', followed by the subtext 'Sign in to manage bookings, profile and tickets.' Below this are two input fields: 'Email Address' and 'Password'. A prominent green button labeled 'Login to Account' is at the bottom, with a link 'Don't have an account? Register' underneath it. The background is a blurred view of the main application interface.

Class Diagram



5.Implementation Details

The implementation of the Airline Reservation System involves developing the software using appropriate programming languages, databases, and tools to manage flight bookings and passenger information efficiently.

5.1 Software and Hardware Specification

Software Requirements:

- Forntend:HTML,CSS,JavaScript,React,MySQL.
- Backend:MySQL.
- Database:MySQL
- Web Server:XAMPP
- Browser:Google Chrome,Microsoft Modern web browser.

Hardware Requirements:

- Minimum 8GB RAM server.
- Google Cloud.
- Client device:PC or Laptop.

6. Testing

6.1 Test Plan

The implementation of the Airline Reservation System involves developing the software using appropriate programming languages, databases, and tools to manage flight bookings and passenger information efficiently.

6.2 Black Box Testing/Data Validation

- Check if the system shows correct results when searching for flights.
- Verify that ticket booking works properly when valid passenger details are entered.
- Test the payment process to ensure successful transactions.

6.3 Test Cases

Login Test Case – Verify that users can log in with valid username and password.

Invalid Login Test Case – Check that the system shows an error message for incorrect login details.

Flight Search Test Case – Verify that users can search flights by entering source, destination, and date.

Flight Availability Test Case – Check whether the system correctly displays available flights.

Ticket Booking Test Case – Verify that passengers can enter their details and book tickets successfully.

Payment Test Case – Ensure that the system processes payments correctly and confirms the booking.

Ticket Cancellation Test Case – Check that users can cancel booked tickets and the seat availability is updated.

View Booking Test Case – Verify that users can view their booking details using booking ID or login account.

Case Data Validation Test – Ensure the system shows an error for invalid or incomplete input details.

6.4 White Box Testing/Functional Validation

- Tests the internal code and program logic of the Airline Reservation System.
- Verifies functions such as login validation, flight search logic, seat availability checking, and ticket booking process.
- Checks whether conditions, loops, and database queries work correctly.
- Ensures that the system functions correctly and processes data accurately without errors.

6.5 Test Results

The testing of the Airline Reservation System shows that all major modules such as login, flight search, ticket booking, payment, and ticket cancellation are working correctly. The system successfully processes user requests, stores booking details in the database, and updates seat availability when a ticket is booked or cancelled. Overall, the system performs efficiently and meets the required functional requirements.

7.Conclusion and Recommendations

Conclusion

The Airline Reservation System provides an efficient and reliable way to manage flight bookings and passenger information. It simplifies the process of searching flights, booking tickets, making payments, and cancelling reservations. The system reduces manual work, saves time, and improves accuracy in managing airline operations. By using this system, passengers can easily access flight information and make reservations online, while airline staff can manage schedules and bookings more effectively. Overall, the Airline Reservation System improves service quality, increases operational efficiency, and provides a convenient travel booking experience.

Recommendation

- **Improve Security:** Implement strong security measures to protect passenger data and payment information.
- **User-Friendly Interface:** Design a simple and easy-to-use interface for passengers and administrators.
- **Online Payment Integration:** Support multiple secure payment methods for convenient transactions.
- **Mobile Application Support:** Develop a mobile app so users can book tickets easily from smartphones.

8.Future Scope

The future scope of the Airline Reservation System includes many improvements and new technologies that can make the system more advanced and user-friendly. In the future, the system can include mobile applications so passengers can easily book tickets from their smartphones. It can also integrate secure online payment gateways to support multiple payment options such as credit cards, debit cards, and digital wallets.

The system can be enhanced with artificial intelligence (AI) to recommend the best flights, prices, and travel options to passengers. Additional features such as real-time flight status, automatic notifications, and online check-in can also be added to improve passenger convenience

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